① Python source and its srcML, read by E1 and E2

Python, lines 10–21

```
10  for h in range(hops):
11      a, b = 1 + 2 * h, 2 + 2 * h
12      qc.h(q[a])
13      qc.cx(q[a], q[b])
14      qc.cx(q[src], q[a])
15      qc.h(q[src])
16      qc.measure(q[src], c[2 * h])
17      qc.measure(q[a], c[2 * h + 1])
18      with qc.if_test((c[2 * h + 1], 1)):
19          qc.x(q[b])
20      with qc.if_test((c[2 * h], 1)):
21          qc.z(q[b])
```

Read by the extensions

`cx`	gate, from the role table
`q[src]`	qubit argument, evaluated in each iteration
`c[2*h+1]`	classical bit, measured and then tested
`if_test`	condition of the block it opens
`14`	source line, from `pos:start`

srcML of lines 14 and 18–19, with positions

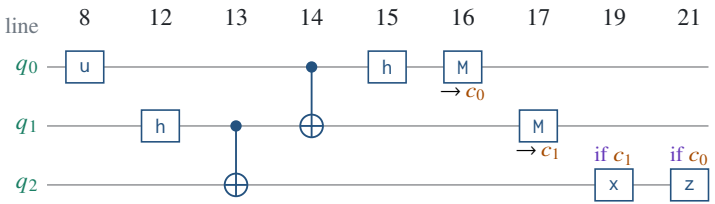
```
<expr_stmt pos:start="14:5"><expr><call><name><name>qc</name><operator>
    ↪ .</operator><name>cx</name></name><argument_list>(<argument><expr>
    ↪ <name><name>q</name><index>[<expr><name>src</name></expr>]</index>
    ↪ </name></expr></argument>, <argument><expr><name><name>q</name><index>[
    ↪ <expr><name>a</name></expr>]</index></name></expr></argument>)
    ↪ </argument_list></call></expr></expr_stmt>
```

```
<with pos:start="18:5">with <expr><call><name><name>qc</name><operator>
    ↪ .</operator><name>if_test</name></name><argument_list>(<argument><expr>
    ↪ <tuple>(<expr><name><name>c</name><index>[<expr><literal type="number">
    ↪ 2</literal><operator>*</operator><name>h</name><operator>+
    ↪ </operator><literal type="number">1</literal></expr>]</index></name>
    ↪ </expr>, <expr><literal type="number">1</literal></expr>)</tuple>
    ↪ </expr></argument>)</argument_list></call></expr><block>:
    ↪ <block_content>
        <expr_stmt><expr><call><name><name>qc</name><operator>.
            ↪ </operator><name>x</name></name><argument_list>(<argument>
            ↪ <expr><name><name>q</name><index>[<expr><name>b</name></expr>]
            ↪ </index></name></expr></argument>)</argument_list></call>
            ↪ </expr></expr_stmt>
</block_content></block></with>
```

Quantum markup of line 14, written back by `qiskit2srcml`

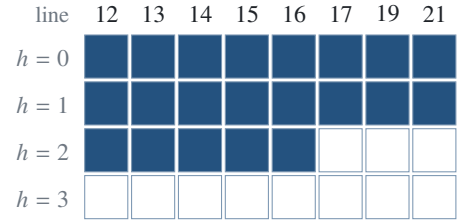
```
<call qu:role="gate" qu:gate="cx" qu:ops="4" qu:qubits="0-7">
    ↪ <argument qu:arg="control">q[src]</argument>,
    ↪ <argument qu:arg="target">q[a]</argument> . . . </call>
```

② E2: trace of line 8 and iteration $h = 0$, as a circuit



Iterations 1 to 3 repeat it on the next qubits. A CNOT, line 14, can change its control q_0 as well as its target.

③ E3: slice of `q[4]` by line and iteration



line 8 in the slice, line 23 not **22** of 34 operations